Customer Churn Prediction Using Machine Learning Classification Algorithms

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Abstract—Retaining existing subscribers costs telecom operators far less than winning new ones, so identifying customers who are likely to leave is a practical priority. This paper studies churn prediction on the IBM Telco Customer Churn dataset, which holds 7,043 customers and 20 usable attributes. After cleaning the data and one-hot encoding the categorical attributes into 30 input features, five classifiers, namely Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors and Support Vector Machine, were trained on a stratified 80:20 split and compared using accuracy, precision, recall, F1 score and area under the ROC curve. Class weighting was used to offset the fact that only about one customer in four churns. Random Forest achieved the highest accuracy (0.7637) and F1 score (0.6194), whereas Logistic Regression gave the highest recall (0.7861) and area under the curve (0.8414). K-Nearest Neighbors, which was trained without class weighting, performed worst. Exploratory analysis and feature importance show that short tenure, month-to-month contracts, fiber optic service, electronic check payment and high monthly charges are the strongest indicators of churn. The results suggest that a simple and interpretable model can compete with an ensemble on this data, and that the final choice should depend on the relative cost of missed churners and false alarms.

Keywords—customer churn, telecommunications, machine learning, classification, Random Forest, class imbalance

I. INTRODUCTION

Telecommunication markets are close to saturation, and a subscriber who is unhappy with a plan can move to a competitor with very little effort. For an operator, every lost subscriber means lost recurring revenue, while replacing that subscriber usually requires marketing and acquisition spending. Predicting which customers are about to leave, so that a retention offer can be targeted at them, is therefore a common data mining application. The task is naturally posed as binary classification: each customer record is labeled as churned or retained, and a model learns the pattern that separates the two groups.

Two difficulties make churn prediction harder than a typical classification exercise. First, churners are a minority, so a model can reach a respectable accuracy simply by favoring the majority class while missing most of the customers that matter. Second, the attributes are a mixture of numeric billing figures and many categorical service options, which must be encoded before most algorithms can use them. This paper addresses both points with a reproducible Python pipeline and a side-by-side comparison of five well-known classifiers.

The main contributions are: (i) a documented preprocessing procedure for the IBM Telco dataset; (ii) a comparison of five classifiers using accuracy, precision, recall, F1 score, confusion matrices and ROC analysis; (iii) an exploratory and feature importance study that identifies the customer characteristics most associated with churn; and (iv) a discussion of how the choice of model depends on business cost. The overall workflow is summarized in Fig. 1. The remainder of the paper reviews related work, describes the data and method, reports the results and concludes.

II. RELATED WORK

Ahmad et al. [1] built a churn prediction model for a large telecom operator on a big data platform and compared tree-based ensemble methods, showing that engineered features from customer activity matter as much as the choice of algorithm. Burez and Van den Poel [2] examined how class imbalance affects churn models and compared sampling and cost-sensitive strategies, including weighted random forests. Synthetic oversampling methods such as SMOTE [3] are another widely used remedy. In this work, class weighting is used instead of resampling because it keeps the original records unchanged and is available directly in the chosen library.

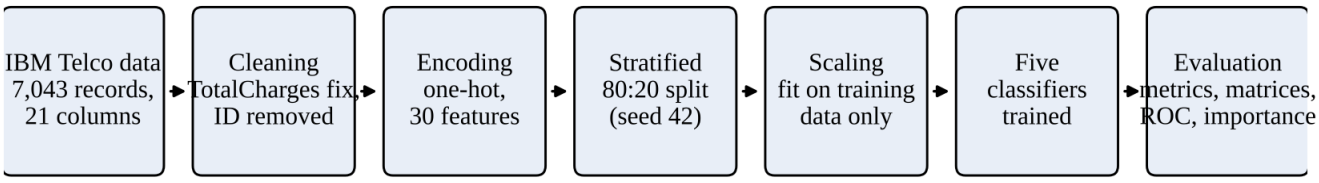


Fig. 1. Workflow of the churn prediction pipeline used in this study.

III. DATASET AND PREPROCESSING

A. Dataset Description

The experiments use the Telco Customer Churn dataset published by IBM [4]. It contains 7,043 customer records and 21 columns. The attributes fall into three groups: demographic details (gender, senior citizen status, partner and dependents), account information (tenure in months, contract type, paperless billing, payment method, monthly charges and total charges) and subscribed services (phone, multiple lines, internet service, online security, online backup, device protection, technical support, streaming television and

streaming movies). The target column records whether the customer left. A total of 1,869 customers churned and 5,174 stayed, so the churn rate is 26.54 percent, as shown in Fig. 2(a). Table I lists the attributes by group.

TABLE I
ATTRIBUTES OF THE TELCO CUSTOMER CHURN DATASET

Group	Attributes
Demographic (4)	gender, SeniorCitizen, Partner, Dependents
Account (6)	tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
Services (9)	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Target (1)	Churn (Yes or No)
Identifier (1)	customerID (removed before modeling)

B. *Data Cleaning and Encoding*

The following steps were applied in order. (1) The data were checked for duplicate rows, and none were found. (2) The TotalCharges column is stored as text, so it was converted to a numeric type. Eleven entries were blank, and all of them belong to customers whose tenure is zero, meaning that no bill had been issued yet; these values were therefore set to zero. (3) The customerID column was removed because it only identifies a record and carries no predictive information. (4) The target was mapped to numbers, with No as 0 and Yes as 1. (5) The remaining categorical columns were converted with one-hot encoding, dropping the first category of each column to avoid redundant variables. This produced 30 input features.

The data were split into 80 percent for training (5,634 records) and 20 percent for testing (1,409 records). Stratified sampling with a fixed random seed of 42 kept the churn proportion the same in both parts. Standardization to zero mean and unit variance was fitted on the training set only and then applied to the test set, which avoids information leaking from the test data. Scaled features were used for Logistic Regression, K-Nearest Neighbors and the Support Vector Machine, which are sensitive to feature magnitude, while the Decision Tree and Random Forest used the unscaled features.

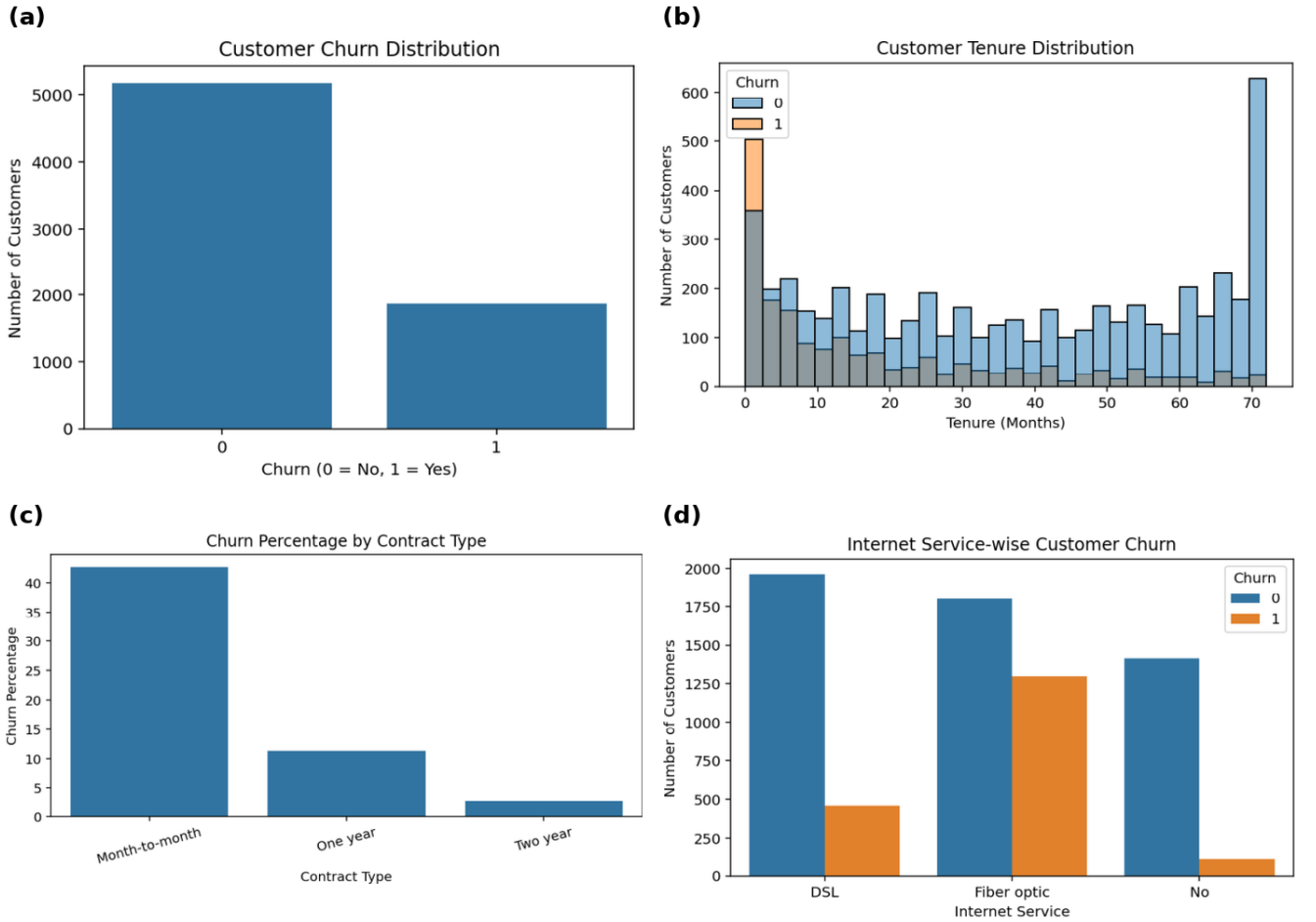


Fig. 2. Exploratory analysis of the churn data: (a) class distribution, (b) tenure distribution by churn status, (c) churn percentage by contract type, (d) churn by internet service type.

C. *Exploratory Analysis*

Exploratory plots were produced before modeling, and the main findings are summarized in Fig. 2. Contract type has the strongest visible effect: 42.71 percent of month-to-month customers churned, against 11.27 percent for one-year and only 2.83 percent for two-year contracts (Fig. 2(c)). Tenure shows a similar pattern. Churners had an average tenure of 18.0 months, compared with 37.6 months for retained customers, and Fig. 2(b) shows that churn is concentrated among customers in their first few months. Churners also paid more, with a mean monthly charge of 74.44 against 61.27 for retained customers.

Among internet customers, fiber optic users churned far more often (1,297 of 3,096, or 41.9 percent) than DSL users (459 of 2,421, or 19.0 percent), and customers with no

internet service rarely left (113 of 1,526) as in Fig. 2(d). Payment by electronic check was also associated with a high churn rate of 45.29 percent, while the other three payment methods ranged from 15.24 to 19.11 percent. Senior citizens churned at 41.7 percent compared with 23.6 percent for other customers, and customers without online security or technical support churned at roughly 42 percent compared with roughly 15 percent for those who had these services. Gender had almost no effect: 26.9 percent of women and 26.2 percent of men churned (Fig. 3(b)).

The box plot in Fig. 3(a) adds to the billing picture. The median monthly charge of churned customers is roughly 80, against roughly 65 for retained customers, and even the lower quartile of the churned group lies far above that of the retained group. Retained customers are spread over the whole price range, including many low-cost plans, whereas churners are concentrated in higher-priced plans. Together, these views suggest that new, high-paying customers on flexible contracts form the highest-risk segment.

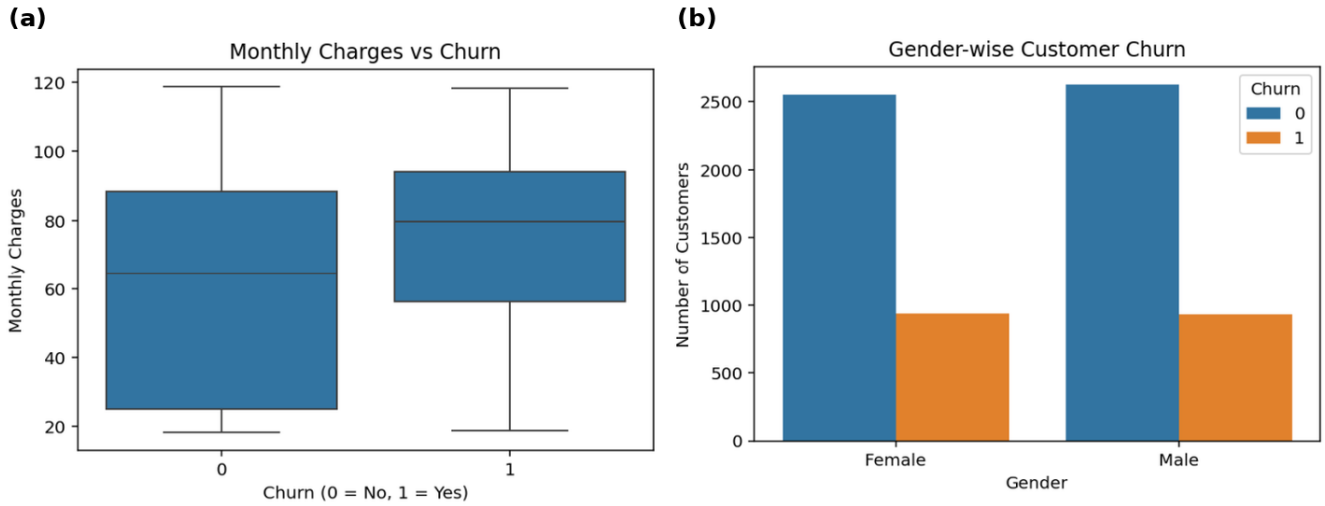


Fig. 3. Monthly charges and gender versus churn: (a) distribution of monthly charges for retained (0) and churned (1) customers, (b) number of customers by gender and churn status.

IV. Methodology

A. Classification Algorithms

All experiments were run in Python 3.13 using the Spyder environment, with pandas for data handling, scikit-learn [5] for modeling and metrics, and matplotlib and seaborn for plots. Five classifiers were trained. Every model that supports it was given balanced class weights, so that errors on the churn class are penalized in proportion to its rarity. The settings are listed in Table II.

TABLE II
Settings Used for Each Classifier

Model	Key settings	Scaled input
Logistic Regression	maximum 1,000 iterations, balanced class weights	Yes
Decision Tree	entropy criterion, maximum depth 5, balanced class weights	No
Random Forest	100 trees, maximum depth 10, balanced class weights	No
KNN	5 neighbors, Euclidean distance, no class weights	Yes
SVM	RBF kernel, probability estimates on, balanced class weights	Yes

All models used a fixed random seed of 42 where applicable.

Logistic Regression [6] models the log-odds of churn as a linear function of the features. It was trained on scaled data with a limit of 1,000 iterations. The Decision Tree [7] splits the data repeatedly on the attribute that gives the largest information gain; the entropy criterion was used with a maximum depth of five to prevent overfitting and keep the rules readable. The Random Forest [8] averages many decorrelated trees built on bootstrap samples; 100 trees with a maximum depth of ten were used. K-Nearest Neighbors [9] labels a customer by a vote among the five closest training records in the scaled feature space; this implementation offers no class weighting. The Support Vector Machine [10] finds a maximum-margin boundary, and a radial basis function kernel was used on scaled data with probability estimates enabled so that a ROC curve could be drawn.

B. Evaluation Metrics

Because accuracy alone can hide poor performance on a minority class, models were judged using the confusion matrix counts of true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN). Writing P for precision and R for recall, the metrics are:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (1)$$

$$P = TP / (TP + FP) \quad (2)$$

$$R = TP / (TP + FN) \quad (3)$$

$$F1 = 2PR / (P + R) \quad (4)$$

Recall measures the share of real churners that are caught, and precision measures the share of flagged customers who really would have left. The F1 score balances the two. The area under the ROC curve (AUC) [11] summarizes how well a model ranks churners above non-churners across all decision thresholds, independent of any single cutoff.

V. Results and Discussion

A. Overall Performance

Table III lists the scores of all five models on the 1,409 test records. Random Forest obtained the best accuracy (0.7637), precision (0.5409) and F1 score (0.6194). Logistic Regression obtained the best recall (0.7861) and AUC

(0.8414). The three leading models are very close in F1 score, with Logistic Regression at 0.6164, the Support Vector Machine at 0.6172 and Random Forest at 0.6194, so the gap between them is small and should not be over-interpreted on a single test split. The Decision Tree was slightly weaker, with an accuracy of 0.7246. K-Nearest Neighbors was clearly the weakest on the minority class, with a recall of only 0.5000 and an F1 score of 0.5123.

TABLE III
PERFORMANCE OF THE FIVE CLASSIFIERS ON THE TEST SET

Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression	0.7402	0.5069	**0.7861**	0.6164	**0.8414**
Decision Tree	0.7246	0.4882	0.7754	0.5992	0.8279
Random Forest	**0.7637**	**0.5409**	0.7246	**0.6194**	0.8400
KNN	0.7473	0.5253	0.5000	0.5123	0.7718
SVM	0.7473	0.5162	0.7674	0.6172	0.8185

Best value in each column is in bold. KNN: K-Nearest Neighbors; SVM: Support Vector Machine.

Table V looks at both classes separately. Random Forest has the best score on the majority class (F1 of 0.83 and recall of 0.78) and the best macro and weighted averages (0.72 and 0.77). K-Nearest Neighbors reaches the highest recall on retained customers (0.84), but this comes at the expense of the churn class, where its F1 score drops to 0.51. Logistic Regression, Random Forest and the Support Vector Machine are tied at about 0.62 on the churn class. Looking at the churn class and the macro average, rather than accuracy alone, therefore gives a more honest view of model quality on this imbalanced task.

B. Confusion Matrix Analysis

The confusion matrices in Table IV show where each model makes its mistakes, and Fig. 4 displays them graphically. Logistic Regression detected 294 of the 374 churners in the test set and missed 80, whereas Random Forest detected 271 and missed 103. In exchange, Random Forest raised 230 false alarms against 286 for Logistic Regression. The Support Vector Machine and the Decision Tree fall between these two in missed churners, with 87 and 84 respectively. K-Nearest Neighbors produced the fewest false alarms (169) but missed 187 churners, which is half of all customers who left. This is consistent with its lack of class weighting: the majority class dominates the neighborhood vote.

TABLE IV
CONFUSION MATRIX COUNTS ON THE TEST SET

Model	TN	FP	FN	TP
Logistic Regression	749	286	80	294
Decision Tree	731	304	84	290
Random Forest	805	230	103	271
KNN	866	169	187	187
SVM	766	269	87	287

Positive class: churn. TN: true negatives; FP: false positives; FN: false negatives; TP: true positives.

TABLE V
CLASS-WISE F1 SCORE AND SUMMARY AVERAGES

Model	Recall (0)	F1 (0)	F1 (1)	Macro F1	Weighted F1
Logistic Regression	0.72	0.80	0.62	0.71	0.75
Decision Tree	0.71	0.79	0.60	0.69	0.74
Random Forest	0.78	0.83	0.62	0.72	0.77
KNN	0.84	0.83	0.51	0.67	0.75
SVM	0.74	0.81	0.62	0.71	0.76

Class 0: retained customers; class 1: churned customers. Values from the classification reports, rounded to two decimals.

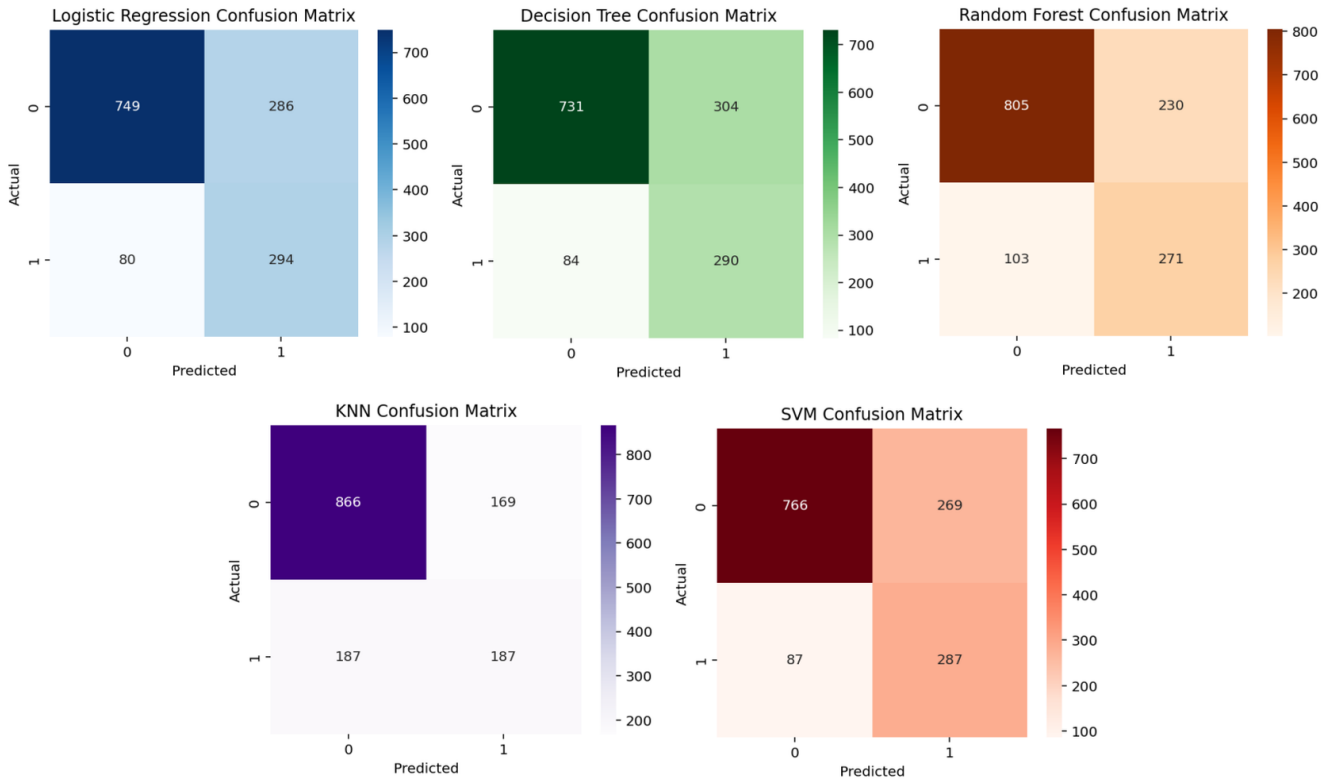


Fig. 4. Confusion matrices of the five classifiers on the test set. First row: Logistic Regression, Decision Tree and Random Forest. Second row: K-Nearest Neighbors and Support Vector Machine. Rows show actual classes and columns show predicted classes (0 = retained, 1 = churned).

C. ROC Analysis

Fig. 5 compares the ROC curves. The curves of Logistic Regression (AUC 0.8414), Random Forest (0.8400), the Decision Tree (0.8279) and the Support Vector Machine (0.8185) lie close together and well above the random-guess diagonal, which indicates that all four rank customers by churn risk reasonably well. The K-Nearest Neighbors curve sits visibly lower (0.7718), confirming its weaker discrimination. The near tie between a linear model and a forest suggests that, with these features, most of the usable signal is already captured by additive effects.

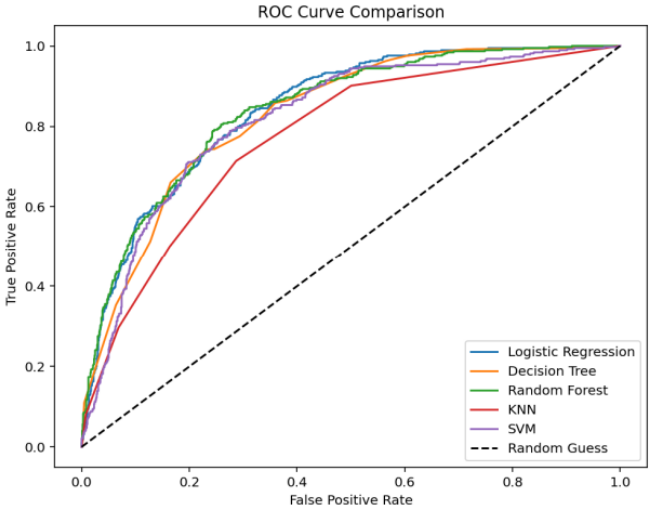


Fig. 5. ROC curves of the five classifiers; the dashed line shows random guessing.

D. Feature Importance

Table VI gives the eight highest Random Forest importance scores. Tenure, total charges and monthly charges are the top three features and together account for about 42 percent of the total importance. Contract type, fiber optic service and electronic check payment follow. These rankings agree with the exploratory findings: long contracts and long tenure protect against churn, while expensive fiber optic plans paid by electronic check are risky. Tenure and total charges are strongly related, since total charges grow with time, so their individual scores should be read with some caution.

TABLE VI
TOP EIGHT RANDOM FOREST FEATURE IMPORTANCES

Feature	Importance
tenure	0.1790
TotalCharges	0.1353
MonthlyCharges	0.1099
Contract (two year)	0.0961
InternetService (fiber optic)	0.0671
PaymentMethod (electronic check)	0.0493
Contract (one year)	0.0417
OnlineSecurity (yes)	0.0396

E. Practical Implications and Limitations

No single model wins on every measure, so the choice should follow the economics of retention. If a retention offer is cheap compared with the revenue lost when a customer leaves, recall matters most, and Logistic Regression is attractive because it catches the most churners, is fast to train and is easy to explain. If the offer is expensive, the extra precision and accuracy of Random Forest reduce wasted offers. To make the contrast concrete, Logistic Regression flags 580 customers in order to reach 294 churners, while Random Forest flags 501 customers to reach 271. For all models, precision stays near 0.5, so about half of the flagged customers would not have left; this is the price paid for class weighting and could be reduced by tuning the decision threshold.

Several limitations apply. The results come from one random split rather than repeated cross-validation, so small differences between models may not be statistically meaningful. Hyperparameters were set by hand and not tuned by search. The dataset is a single sample of about seven thousand customers with no information on customer complaints or usage over time, so the findings may not transfer directly to another operator.

VI. CONCLUSION AND FUTURE WORK

This paper compared Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors and Support Vector Machine on the IBM Telco churn data after a documented cleaning and encoding procedure. Random Forest gave the best accuracy (0.7637) and F1 score (0.6194), Logistic Regression gave the best recall (0.7861) and AUC (0.8414), and K-Nearest Neighbors was the weakest. Short tenure, month-to-month contracts, fiber optic service, electronic check payment and high monthly charges emerged as the clearest churn signals. Future work will use repeated cross-validation and grid search, test gradient boosting and resampling methods such as SMOTE [3], tune the decision threshold against a cost model, and explain individual predictions with model-agnostic tools.

REFERENCES

[1] A. Ahmad, A. Jafar, and K. Aljoumaa, “Customer churn prediction in telecom using machine learning in big data platform,” J. Big Data, vol. 6, Art. no. 28, 2019.

[2] J. Burez and D. Van den Poel, “Handling class imbalance in customer churn prediction,” Expert Syst. Appl., vol. 36, no. 3, pp. 4626–4636, 2009.

[3] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique,” J. Artif. Intell. Res., vol. 16, pp. 321–357, 2002.

[4] IBM, “Telco customer churn,” sample dataset, IBM Business Analytics Community. [Online]. Available: https://www.kaggle.com/datasets/blastchar/telco-customer-churn

[5] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.

[6] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, Applied Logistic Regression, 3rd ed. Hoboken, NJ, USA: Wiley, 2013.

[7] J. R. Quinlan, “Induction of decision trees,” Mach. Learn., vol. 1, no. 1, pp. 81–106, 1986.

[8] L. Breiman, “Random forests,” Mach. Learn., vol. 45, no. 1, pp. 5–32, 2001.

[9] T. Cover and P. Hart, “Nearest neighbor pattern classification,” IEEE Trans. Inf. Theory, vol. 13, no. 1, pp. 21–27, Jan. 1967.

[10] C. Cortes and V. Vapnik, “Support-vector networks,” Mach. Learn., vol. 20, no. 3, pp. 273–297, 1995.

[11] T. Fawcett, “An introduction to ROC analysis,” Pattern Recognit. Lett., vol. 27, no. 8, pp. 861–874, 2006.